EXPERIMENT NO: 01

Exploring Matplotlib, Seaborn, and Plotly

AIM:

To perform data exploration, cleaning, and visualization on a large real-world dataset (Diamond dataset) using Matplotlib, Seaborn, and Plotly, and to derive analytical insights from different graph types.

LIBRARIES:

- Pandas
- Matplotlib
- Seaborn
- Plotly Express

PROCEDURE:

1. Read the dataset from the URL using pandas.read_csv().

2. Check missing values with df.isnull().sum() and drop null rows.

3. Remove irrelevant columns like 'depth' and 'table'.

4. Take a sub-sample of 3,000 rows for fast Plotly interactive plots.

5. Generate Bar Plot, Scatter Plot, Pie Chart, Histogram, and Box Plot using Matplotlib, Seaborn, and Plotly.

6. Derive insights from each generated chart.

PYTHON CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

# 1. Dataset Loading
url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/diamonds.csv"
df = pd.read_csv(url)

# 2. Data Exploration & Inspection
print("--- DATA EXPLORATION ---")
print("Column Names:\n", df.columns.tolist())
print("Dataset Shape:", df.shape)
print("\nMissing Values Per Column:\n", df.isnull().sum())

# 3. Data Cleaning & Sub-sampling
```

```
df_cleaned = df.dropna().drop(columns=['depth', 'table'])
df_sample = df_cleaned.sample(n=3000, random_state=42)

print("\n--- GENERATING PLOTS ---")

# Plot 1: Bar Plot (Seaborn) - Average Price by Cut
plt.figure(figsize=(8, 5))
sns.barplot(data=df_cleaned, x='cut', y='price', palette='viridis', errorbar=None)
plt.title('Average Diamond Price by Cut Quality')
plt.xlabel('Cut')
plt.ylabel('Average Price ($)')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()

# Plot 2: Interactive Scatter Plot (Plotly) - Carat vs Price
fig_scatter = px.scatter(
    df_sample,
    x='carat',
    y='price',
    color='clarity',
    title='Diamond Carat vs Price (Sampled Interactive Scatter)',
    labels={'carat': 'Carat Weight', 'price': 'Price ($)'},
    opacity=0.7
)
fig_scatter.show()

# Plot 3: Pie Chart (Matplotlib) - Proportion of Cuts
cut_counts = df_cleaned['cut'].value_counts()
plt.figure(figsize=(7, 7))
plt.pie(
    cut_counts,
    labels=cut_counts.index,
    autopct='%1.1f%%',
    startangle=140,
    colors=sns.color_palette('pastel')
)
plt.title('Proportion of Diamond Cuts Available')
plt.tight_layout()
plt.show()

# Plot 4: Histogram (Seaborn) - Price Distribution
plt.figure(figsize=(8, 5))
sns.histplot(data=df_cleaned, x='price', kde=True, bins=50, color='Crimson')
plt.title('Distribution of Diamond Prices')
plt.xlabel('Price ($)')
plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
```

```
# Plot 5: Interactive Box Plot (Plotly) - Price Across Colors
fig_box = px.box(
    df_sample,
    x='color',
    y='price',
    color='color',
    title='Price Distribution Across Diamond Colors (Box Plot)',
    category_orders={'color': ['D', 'E', 'F', 'G', 'H', 'I', 'J']}
)
fig_box.show()
```

VISUALIZATION & ANALYTICAL INSIGHTS:

1. Ideal Cuts vs Fair Cuts: Ideal cuts show a lower average price than 'Fair' cuts because they tend to be smaller in carat weight.

2. Price vs Carat Weight: Price increases exponentially with carat weight, with higher clarity grades sitting at the top price levels.

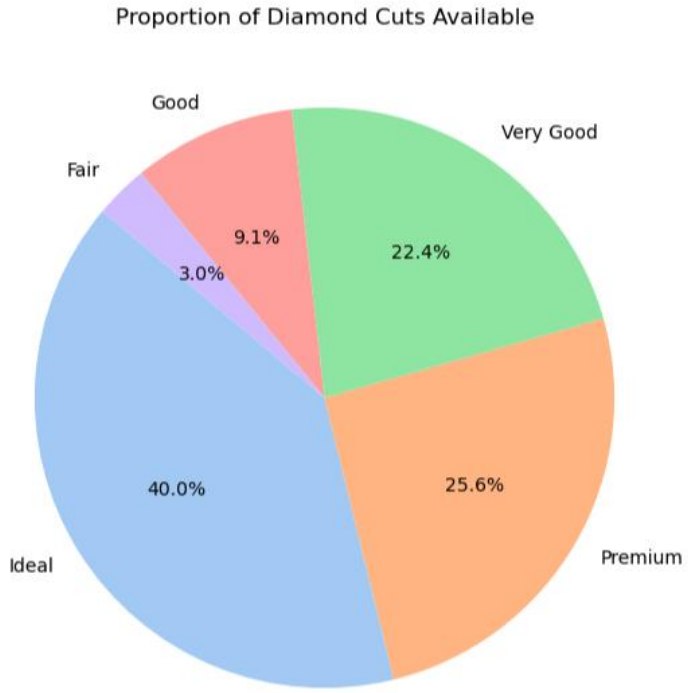
3. Cut Proportions: Over 40% of the inventory consists of 'Ideal' cuts, while 'Fair' cuts make up only about 3%.

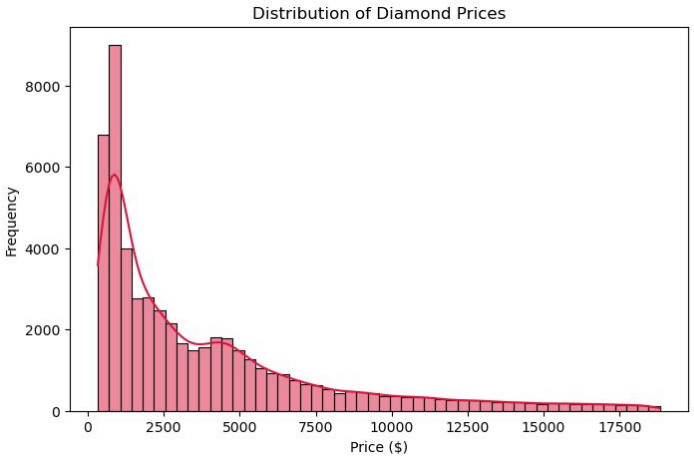
4. Price Distribution: Prices are strongly right-skewed, peaking heavily under $2,000 with a long thin tail extending to $18,000.

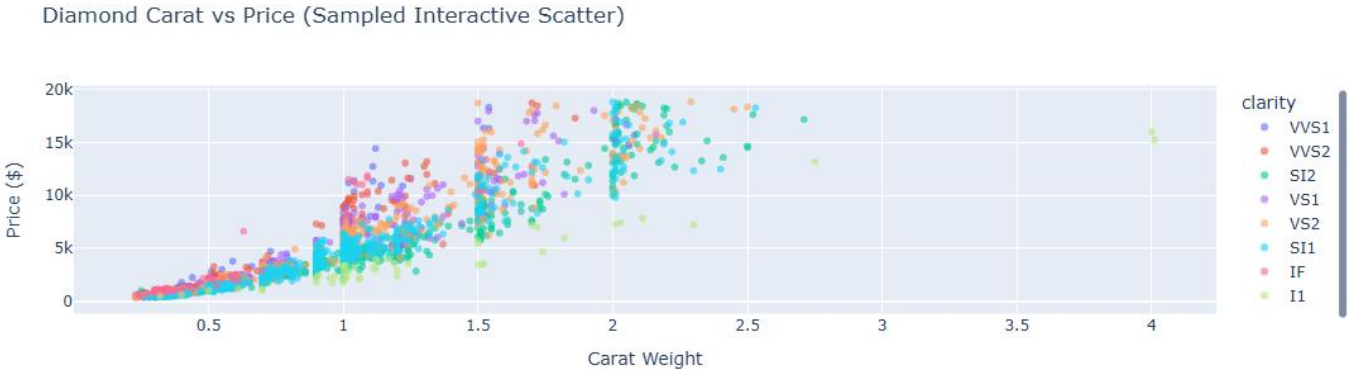
5. Color vs Price Trend: Median price and spread increase as color grade drops (D to J) due to lower color grade diamonds being larger on average.

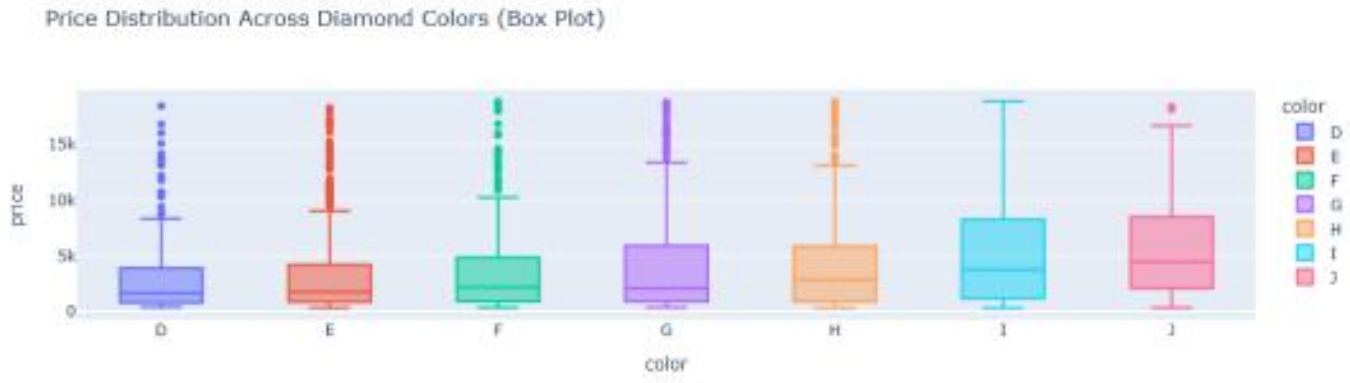
OUTPUT:


Average Diamond Price by Cut Quality
Average Price ($)
0
1000
2000
3000
4000
Ideal
Premium
Good
Very Good
Fair
Cut


Proportion of Diamond Cuts Available
Good
Very Good
Fair
9.1%
22.4%
3.0%
40.0%
25.6%
Ideal
Premium







RESULT:

Data exploration, cleaning, and visualization were successfully completed on the Diamond dataset using Matplotlib, Seaborn, and Plotly, and key analytical insights were derived.